
Trustworthy and Efficient Map-free LiDAR Localization via Scan-Pose Alignment and flow-matching

Anonymous Author(s)

Affiliation

Address

email

Abstract

Reliable map-free LiDAR relocalization requires not only accurate and efficient pose estimation but also trustworthy confidence estimation for detecting localization failures. However, estimating reliability to ensure trustworthy LiDAR relocalization remains largely unexplored. Also, existing LiDAR pose regression methods suffer from accuracy and efficiency trade-off. To address this limitation, we propose **SPAR**, a model-agnostic confidence estimator that reformulates localization reliability as scan-pose alignment. Given a query scan and a predicted pose, SPAR measures their compatibility in a learned embedding space and explicitly optimizes confidence scores to correlate scan with pose. In addition, we introduce **VeLoc**, an efficient and accurate pose regressor based on flow-matching for efficient pose refinement. Experiments on the Oxford and NCLT datasets demonstrate that the proposed framework achieves a favorable balance between efficiency and accuracy for LiDAR relocalization while enabling reliable localization failure detection across different localization backbones.

1 Introduction

LiDAR relocalization, defined as estimating 6-DoF pose from a LiDAR scan, is crucial for computer vision and robotics tasks [1, 2]. Existing approaches can be categorized into map-based retrieval methods and map-free implicit methods [3]. Map-based methods maintain a scan database for retrieval [4] and result in burdensome map maintenance and storage overheads [5]. Consequently, implicit localization methods, which can be categorized into SCR [6] and APR [7], are increasingly preferred [8].

Despite SCR achieving higher precision via correspondence estimation and RANSAC [9], it suffers from lower computational efficiency. Conversely, APR directly regresses poses, offering a simpler inference pipeline, lower runtime overhead, and better compatibility with other end-to-end tasks. However, current methods suffer from two limitations as follows.

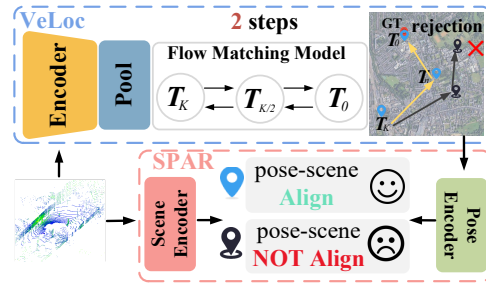


Figure 1: **Model overview.** Our method **VeLoc** outperforms the SOTA APR baseline and achieves competitive performance compared to the SOTA SCR baseline with faster speed. Also, our **SPAR** module effectively rejects outliers.

First, current methods largely lack trustworthy evaluation of the reliability of pose predictions. We analyze the error distribution of leading methods like [6] and find out that pose estimations are highly accurate 95% of the time with errors under 4.8 meters. However, in the worst 1% of cases, the error drastically rises to over 11 meters, causing catastrophic deviations. Prior works mainly evaluate performance via average errors and lack such analysis. Such metrics inherently mask the rare but severe failures that are critical to safe autonomy. This issue necessitates an efficient uncertainty estimation method for outlier rejection. However, current uncertainty estimation approaches suffer from practical limitations including theoretical fragility and high computational costs [10–12]. Thus, how to design a model-agnostic confidence estimation method that can effectively reject outliers within only one forward pass becomes an important question.

Second, current methods fall short of achieving a favorable balance between accuracy and efficiency. As shown in Table 1, although DiffLoc and SGLoc achieve impressive accuracy, they suffer from slow inference speed. In contrast, restricting DiffLoc to only one step causes a dramatic performance drop, indicating that diffusion strongly relies on multi-step denoising, which inevitably weakens the efficiency advantage of APR. These observations motivate us to seek a few-step formulation that retains the benefit of refinement for accuracy while preserving a compact and efficient inference pipeline.

Table 1: **Accuracy-efficiency trade-off analysis.** We compare both APR and SCR methods on NCLT dataset.

Methods	Type	Trans. [m]	Rot. [°]	Latency [ms]
DiffLoc[13]	APR	✓1.19	✓2.31	✗34.29
DiffLoc(1-step)[13]	APR	✗8.68	✗7.00	✓9.44
SGLoc[6]	SCR	✓1.83	✗3.54	✗62.33

Based on the observations above, we fully analyze the current localization and uncertainty estimation pipelines. For uncertainty estimation, inspired by VLM [14, 15], which explicitly learn a joint embedding space to measure cross-modal compatibility via symmetric contrastive learning, we reformulate confidence estimation as measuring the compatibility between the query scan and the predicted pose in a learned embedding space. Unlike heuristic methods that derive uncertainty without direct supervision or stochastic sampling from empirical statistics, our formulation yields an uncertainty estimation that is grounded in scan-pose correspondence. For efficiency and accuracy trade-off, we formulate iterative refinement in relocalization as progressively modifying the current pose estimate by modeling how the pose should gradually update toward its next state. The resulting accuracy-efficiency imbalance motivates us to model a few-step velocity field that better balances them. We therefore formulate iterative refinement as learning a conditional velocity field along a straight correction path, which aligns naturally with flow-matching.

In this paper, we propose SPAR, a plug-and-play uncertainty estimation module that enables robust failure detection, and VeLoc, a flow-matching-based APR framework that achieves efficient and accurate LiDAR relocalization, as shown in Figure 1. SPAR is designed as a dual-stream architecture with a queue of poses for negative samples. A sparse voxel-based GeoNet backbone with multi-scale RFF encoders is designed to encode scans into high-dimensional embeddings. Additionally, a novel compound loss and a coarse-to-fine strategy are designed for scan-pose alignment and score-error correlation. Furthermore, we propose VeLoc, introducing flow-matching into the pose regression. The input scan is projected into range images and processed using a ViT with DINO weights. Subsequently, a SOAP module is deployed for moving objects removal. The contributions of our work are summarized as follows:

- First, we propose SPAR, a post-processing module based on embedding alignment to estimate the localization uncertainty. Our approach serves as a model-agnostic module that can effectively reject outliers.
- Second, we propose VeLoc, a conditional flow-matching-based APR framework, which achieves competitive localization accuracy while reducing the localization inference latency by $3\times$ compared to the SOTA baselines.
- Third, we design a novel compound loss and a coarse-to-fine strategy tailored for scan-pose alignment. Experiments demonstrate that our design enforces a strong negative correlation between confidence scores and localization errors.

2 Related Work

2.1 Absolute Pose Regression and Generative Refinement

APR directly regresses the 6-DoF pose from the input in an end-to-end manner [16]. Beyond [16] for image localization, some image localization works focus on estimating the uncertainty of the regression results [11, 17–19] simultaneously. Based on [16], PointLoc [20] extends the pipeline to point clouds, employing a PointNet-style backbone [21] to handle unstructured data. Beyond PointLoc, various works have been proposed to improve regression accuracy [7, 20, 22–27]. Subsequent approaches enhance pose regression by incorporating auxiliary mechanisms, such as feature fusion [27], neuro-inspired memory [26], and spatio-temporal constraints [28]. Meanwhile, LiDAR-based odometry such as [29–31] also shares a similar regression pipeline and thus provides helpful insights for APR. Recently, generative models have been explored as refinement mechanisms for regression tasks. Diffusion model was initially proposed as a powerful generative model [32–35]. Grounded in Langevin dynamics, diffusion model effectively mitigates the training instability in GANs [36]. Recent works explore leveraging diffusion models as regressors for discriminative tasks [37, 38]. However, the inherent multiple denoising steps lead to slow inference speed [39, 40]. To address this bottleneck, flow-matching models [41, 42] have emerged as a promising direction, enabling efficient inference by modeling velocity to construct straight-line trajectories [43, 44].

2.2 Uncertainty Estimation

Uncertainty estimation has received increasing attention but remains largely underexplored in LiDAR-based relocalization. Existing image-based methods estimate uncertainty with auxiliary branches [10, 45] or predefined pose-error distributions [11, 18, 46]. Many of these estimates are empirically designed and lack strong theoretical grounding [12]. Generic techniques such as Monte Carlo (MC) dropout and deep ensembles are impractical for deployment, as they require multiple forward passes, and ensembles further require training multiple models. In LiDAR-based relocalization, DiffLoc [13] first uses sample variance as an uncertainty metric. However, variance can be poorly calibrated under tightly clustered samples and requires multiple predictions at inference time. Another branch of work estimates reliability from retrieval confidence. For example, HR-APR [47] uses the similarity score of the Top-1 retrieved result as an uncertainty indicator. This perspective suggests that uncertainty can be inferred from how well a query matches its reference representation. This naturally connects uncertainty estimation to contrastive learning, which has been widely used for cross-modal alignment [48, 49] and large-scale geolocalization [50, 51]. However, although conventional contrastive frameworks [52, 53] excel at discrete discrimination, they are inherently ill-suited for continuous regression. To bridge this gap, recent advances in continuous contrastive learning [54, 55] exploit global ranking to learn representations tailored to regression tasks.

3 Method

3.1 SPAR Module

To estimate the reliability of estimated pose, we propose **SPAR** (Scan-Pose Alignment Reliability Estimator) module, which is designed as a dual-stream architecture to extract features from scene and pose, respectively, with a queue of past poses to expand the negative sample pool. The schematic overview is illustrated in Figure 2.

Scan encoder. The scan encoder consists of a scene feature branch and a geometric feature branch. Relying solely on range images (scene feature branch) is insufficient for distinguishing spatially adjacent scans due to their limited granularity and highly similar appearance. To address this limitation, we introduce a lightweight sparse voxel-based network GeoNet (geometric feature branch) to extract robust structural representations from point cloud voxels. The GeoNet adopts a U-Net structure with skip connections to fuse deep and shallow geometric features. Furthermore, we adopt a coordinate-aware pooling module at the bottleneck to preserve geometric details. Specifically, we explicitly inject the normalized coordinates $\mathbf{c} \in \mathbb{R}^3$ into the feature space before the final global pooling. This augmented tensor is then processed by an MLP to generate a position-aware feature embedding. Finally, we employ a gated fusion module which generates a soft mask to integrate the semantic information from the scene branch and the structural information from the geometry branch.

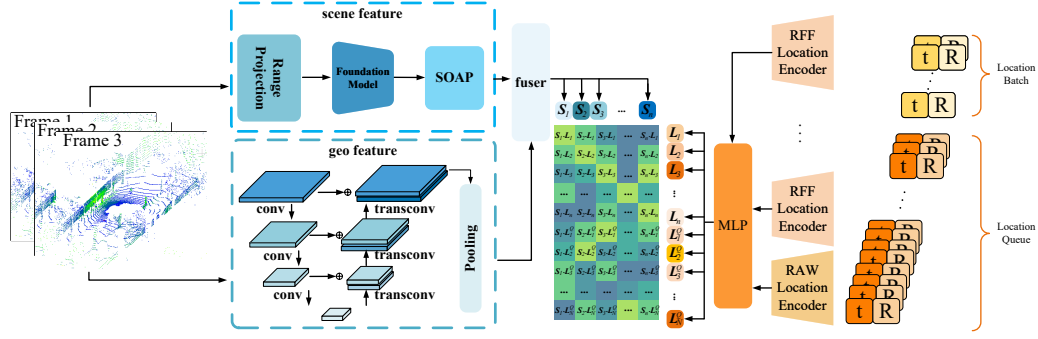


Figure 2: **SPAR module overall architecture.** The framework adopts a dual-stream design. The **scan stream (blue)** extracts and fuses scene feature and sparse voxel-based geometric feature. The **pose stream (orange)** encodes pose data using RFF and raw encoders. A pose memory queue stores past poses to expand the negative sample pool.

Pose encoder. We use MLP architecture as pose encoder and find that standard MLPs typically lack the ability to capture high-frequency variations between different poses [56]. To mitigate this limitation, we leverage Random Fourier Features (RFF) [56] to map the input poses into higher-dimensional embeddings. Specifically, our encoder consists of N location capsules and a raw encoding capsule. For the i -th capsule, the input pose $\mathbf{p} \in \mathbb{R}^6$ is first mapped via Gaussian encoding with a specific frequency scale σ , followed by an MLP to extract parametric features. For σ , we employ a fixed interval schedule from 2^0 to 2^{10} to ensure discrimination for both global and local pose differences. Simultaneously, the raw capsule directly processes the original coordinates to preserve linear information. Finally, all features are concatenated and fused via a projection head. To learn a robust feature representation during training, a sufficiently large negative pool is a standard and essential component in contrastive learning [53]. However, the number of in-batch negatives is inherently constrained by the batch size, which limits sample diversity during training. To address this issue, we employ a pose queue Q to store past samples as additional negatives. At each forward pass, the queued poses are concatenated with the current in-batch samples to enlarge the negative pool. After each iteration, the queue is updated in a first-in-first-out (FIFO) manner, where newly generated negative samples are enqueued while the oldest samples are dequeued.

Training Pipeline SPAR employs a coarse-to-fine training strategy as outlined in Algorithm 1. In each phase, a contrastive loss and a rank loss are employed for embedding alignment and for enforcing a negative correlation between scores and errors, respectively. We adopt the ADD metric [57] to unify the rotation and translation error. See Appendix A for more details. The $\mathcal{L}_{\text{inference}}^{\text{inter}}$ [53, 58] is employed to align scan and pose embeddings in the first phase. We design a novel loss $\mathcal{L}_{\text{rank}}^{\text{inter}}$ that can be formulated as $\mathcal{L}_{\text{rank}}^{\text{inter}} = 1 - \rho(\mathbf{S}_i^-, \mathbf{T}_i^-)$ where $\rho(\cdot, \cdot)$ denotes the Pearson coefficient. $\mathbf{S}_i^-$ and $\mathbf{T}_i^-$ represent the scores and the target scores derived from geometric distances, respectively. Crucially, we exclude the ground truth (GT) sample from the ranking loss calculation to mitigate optimization conflicts in both phases. The second phase refines local discrimination by freezing low-frequency capsules and synthesizing noisy candidates. The $\mathcal{L}_{\text{inference}}^{\text{intra}}$ can be formulated as:

$$\mathcal{L}_{\text{inference}}^{\text{intra}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(s(I_i, P_{i,0}))}{\sum_{k=0}^{K-1} \exp(s(I_i, P_{i,k}))}, \quad (1)$$

where $P_{i,0}$ denotes the ground truth pose, and $P_{i,k}$ represents the k -th perturbed pose for the i -th scan. Simultaneously, $\mathcal{L}_{\text{rank}}^{\text{intra}}$ imposes a pairwise ordering constraint as:

$$\mathcal{L}_{\text{rank}}^{\text{intra}} = -\frac{1}{|\mathcal{K}|} \sum_{(j,k) \in \mathcal{K}} \log \sigma(s_{i,j} - s_{i,k}), \quad (2)$$

where $\sigma(\cdot)$ is the sigmoid function. $\mathcal{K}$ denotes the set of valid index pairs (j, k) where the geometric error between $P_{i,j}$ and $P_{i,k}$ exceeds the threshold τ . $s_{i,j}$ represents the similarity score $s(I_i, P_{i,j})$.

Algorithm 1 Training procedure for SPAR

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1: Input: Image  $\mathbf{I}$ , Pose  $\mathbf{P}$ , Scan  $\mathbf{S} \in \mathcal{D}$ , Encoders  $\Phi_{\text{loc}}, \Phi_{\text{img}}, \Phi_{\text{geo}}$ , Queue  $\mathcal{Q}$ , Similarity matrix  $\mathbf{Z}$ , Relative point-wise error  $\mathbf{E}$ 
2: if Stage == 'inter-scan' then
3:   Freeze  $\Phi_{\text{img}}$ 
4:   for each batch  $(\mathbf{I}, \mathbf{P}, \mathbf{S}) \in \mathcal{D}$  do
5:     Sample  $t \sim \mathcal{U}[0, N - 1]$ ;  $\mathbf{f}_{\text{img}} \leftarrow \Phi_{\text{img}}(\mathbf{I}_t)$ 
6:      $\mathbf{P}_{\text{set}} \leftarrow [\mathbf{P}_t; \mathcal{Q}]$ ; Update  $\mathcal{Q}$ ;
7:      $\mathbf{Z} \leftarrow \text{Sim}(\mathbf{f}_{\text{img}}, \Phi_{\text{loc}}(\mathbf{P}_{\text{set}}))$ 
8:      $\mathbf{Z}^- \leftarrow \mathbf{Z} \setminus \{z_{\text{gt}}\}$ ;  $\mathbf{E}^- \leftarrow \text{ADD}(\mathbf{P}_{\text{set}} \setminus \{\mathbf{P}_t\})$ 
9:      $\mathcal{L} \leftarrow \mathcal{L}_{\text{InfoNCE}}(\mathbf{Z}) + \lambda \mathcal{L}_{\text{Rank}}(\mathbf{Z}^-, \mathbf{E}^-)$ ; Update  $\Phi_{\text{loc}}$ 
10:   end for
11: else
12:   Freeze  $\Phi_{\text{img}}$  and low-freq capsules in  $\Phi_{\text{loc}}$ 
13:   for each batch  $(\mathbf{I}, \mathbf{P}, \mathbf{S}) \in \mathcal{D}$  do
14:      $\mathbf{P}_{\text{neg}} \sim \mathcal{N}(\mathbf{P}_{\text{gt}}, \sigma)$ ;
15:      $\mathbf{f}_{\text{scan}} \leftarrow \text{Fuse}(\Phi_{\text{img}}(\mathbf{I}), \Phi_{\text{geo}}(\mathbf{S}))$ 
16:      $\mathbf{s} \leftarrow \text{Sim}(\mathbf{f}_{\text{scan}}, \Phi_{\text{loc}}(\mathbf{P}_{\text{neg}}))$ ;  $\mathbf{E}_{\text{neg}} \leftarrow \text{ADD}(\mathbf{P}_{\text{neg}})$ 
17:      $\mathcal{L} \leftarrow \mathcal{L}_{\text{InfoNCE}}(\mathbf{s}) + \mathcal{L}_{\text{Rank}}(\mathbf{s}, \mathbf{E}_{\text{neg}})$ ; Update  $\Phi_{\text{loc}}, \Phi_{\text{geo}}$ 
18:   end for
19: end if

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3.2 VeLoc

To realize accurate and efficient localization, we propose VeLoc, following the paradigm of APR methods which consists of a feature learner and a pose regressor as illustrated in Figure 3.

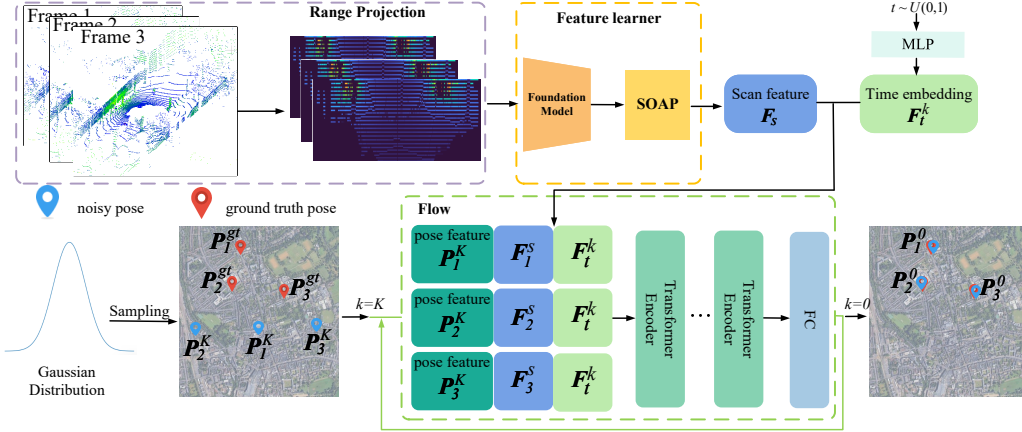


Figure 3: **VeLoc overall architecture.** Input LiDAR frames are first projected into range images. The **feature learner** module comprises a foundation model and a SOAP module. The pose regression task takes the scan features $\mathbf{F}_s$, a shared time embedding $\mathbf{F}_t^k$, and noisy poses $\mathbf{P}_i^K$ sampled from a Gaussian distribution as input. A **Transformer-based flow network** generates the final pose $\mathbf{P}_i^0$.

Feature representation. To leverage the structural regularity and computational efficiency of 2D-conv in unstructured point clouds, we project the point cloud $\mathcal{P} = \{(x_i, y_i, z_i)\}_{i=1}^N$ to range image grid of size $H \times W$. We sort the candidates by range depth and retain the point with the minimum depth when multiple points are projected to the same pixel. Then, we leverage the representation ability of foundation models trained on large-scale datasets following prior works [13, 59]. To bridge the domain gap between range and RGB images, we replace the embedding layer with convolution stem and feed the output to a ViT with DINO parameters. Finally, we incorporate the SOAP module in DiffLoc[13] to suppress artifacts from moving objects[60]. We represent the pose as $y = [t, r] \in \mathbb{R}^6$, where r is the axis-angle rotation obtained by applying the logarithm map to the quaternion. This avoids unit-norm constraints of quaternion and gimbal lock issues of Euler angles. Flow-matching is performed in the Euclidean $\mathbb{R}^6$ space without enforcing the $SO(3)$ geometry, providing an efficient approximation for autonomous driving scenarios. More details are shown in Appendix.A.3

flow-matching pose regressor. We formulate pose regression as a conditional flow-matching problem which constructs a deterministic optimal transport path from a standard Gaussian distribution $x_0 \sim \mathcal{N}(0, I)$ to the target pose x_1 , which is defined as $x_t = (1 - t)x_0 + tx_1$, parameterized by time $t \in [0, 1]$. This linear interpolation defines a straight-line velocity field and its time derivative corresponds to a constant velocity field derived as $u_t(x_t|x_1, x_0) = \frac{d}{dt}x_t = x_1 - x_0$. The network v_θ is trained to regress the target velocity by minimizing the MSE loss

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(x_t, t, c) - (x_1 - x_0)\|^2] \quad (3)$$

where c denotes the query scan embedding. During inference, given a query scan embedding c and sampled noise x_0 , the final pose $\hat{x}_1$ is recovered by numerically integrating the velocity field from $t = 0$ to $t = 1$. In practice, we approximate this integral using numerical ordinary differential equation (ODE) solvers (e.g., Euler method[61]) with discrete time steps.

4 Experiments

4.1 Experiments Setup

We conduct experiments on the NCLT[62] and Oxford [63] datasets. Our method is implemented with MinkowskiEngine[64] and PyTorch. Further training details are provided in the Section A.

4.2 Localization Accuracy Comparison

NCLT dataset. Table 2 summarizes the results on the NCLT dataset. VeLoc achieves the best performance on both metrics against all evaluated baselines. First, VeLoc outperforms LightLoc, reducing translation and rotation error by 30.1% and 18.2%, respectively. Our generative paradigm effectively preserves localization consistency, whereas SCR methods often suffer from feature mismatching in dynamic environments. Second, compared with DiffLoc, the strongest APR baseline, VeLoc achieves an approximately $3 \times$ speedup while reducing the average translation and rotation errors by 47.7% and 27.5%, respectively. Visualizations are provided in Figure 4.

Table 2: **Comparison with the state-of-the-art methods on NCLT dataset.** The mechanism (Mech.) S/A denotes the structure-based localization/absolute pose regression. We highlight the **best** and **second-best** results. **Latency** is measured with a batch size of 1 to simulate real-world usage.

Methods	Mech.	2012-02-12	2012-02-19	2012-03-31	2012-05-26	Average [m/°]	Ranks
SGLoc[6]	S	1.20/3.08	1.20/3.05	1.12/3.28	3.81/4.74	1.83/3.54	4/6
LightLoc[65]	S	0.98 /2.76	0.89 /2.51	0.86 /2.67	3.10/3.26	1.46/2.80	3/3
PoseSOE[7]	A	13.09/8.05	6.16/4.51	5.24/4.56	12.60/7.67	9.27/6.20	8/8
STCLoc[28]	A	4.91/4.34	3.25/3.10	3.75/4.04	8.67/5.23	5.15/4.18	7/7
NIDALoc[26]	A	4.48/3.59	3.14/2.52	3.67/3.46	6.60/4.56	4.47/3.53	6/5
HypLiLoc[27]	A	1.71/3.56	1.68/2.69	1.52/2.90	2.90/3.47	1.95/3.16	5/4
DiffLoc[13]	A	0.99/ 2.40	0.92/ 2.14	0.98/ 2.27	1.88 / 2.43	1.19 / 2.31	2 / 2
VeLoc(Ours)	A	0.82 / 2.44	0.73 / 2.12	0.77 / 2.23	1.78 / 2.39	1.02 / 2.29	1 / 1

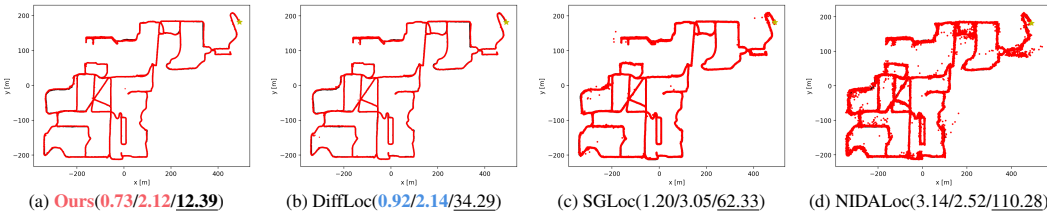


Figure 4: **Visualization of trajectories on the NCLT dataset.** The starting frame is marked by a star. The black and red lines represent the ground truth and the predicted paths, respectively.

Oxford dataset. As detailed in Table 3 and Figure 5, our proposed VeLoc achieves competitive performance against evaluated baselines. First, compared to APR baselines, VeLoc significantly reduces localization errors. Specifically, VeLoc reduces the translation and rotation errors by 7.2% and 22.2%, respectively, with $2 \times$ speedup when compared to APR SOTA, DiffLoc. Second, in

comparison to LightLoc, which ranks first in translation accuracy, VeLoc reduces the rotation error by nearly twofold. While SCR methods excel at position alignment, our regression approach offers superior robustness in estimating rotation. More analyses are shown in Appendix.B.1

Table 3: **Comparison with the baseline methods on Oxford dataset.** The mechanism (Mech.) S/A denotes the structure-based localization/absolute pose regression. We highlight the **best** and **second-best** results. **Latency** is measured with a batch size of 1 to simulate real-world usage.

Methods	Mech.	15-13-06-37	17-13-26-39	17-14-03-00	18-14-14-42	Average [m/°]	Ranks
SGLoc[6]	S	3.01 /1.91	4.07/2.07	3.37 /1.89	2.12 /1.66	3.14 /1.88	2/7
LightLoc[65]	S	2.33 /1.21	3.19 /1.34	3.11 /1.24	2.05 /1.20	2.67 /1.25	1/4
PoseSOE[7]	A	7.59/1.94	10.39/2.08	9.21/2.12	7.27/1.87	8.62/2.00	8/8
STCLoc[28]	A	6.93/1.48	7.55/1.23	7.44/1.24	6.13/1.15	7.01/1.28	7/5
NIDALoc[26]	A	5.45/1.40	7.63/1.56	6.68/1.26	4.80/1.18	6.14/1.35	6/6
HypLiLoc[27]	A	6.88/1.09	6.79/1.29	5.82/0.97	3.45/0.84	5.74/1.05	5/3
DiffLoc[13]	A	3.57/ 0.88	3.65 / 0.68	4.03/ 0.70	2.86/ 0.60	3.53/ 0.72	4/2
VeLoc(Ours)	A	3.30/ 0.92	3.76/ 0.62	3.94/ 0.59	2.66/ 0.55	3.42/ 0.66	3/1

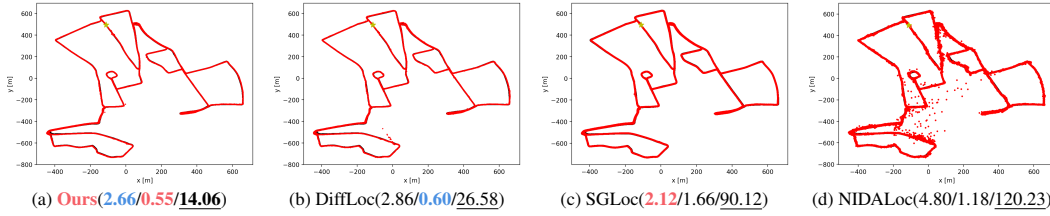


Figure 5: **Visualization of trajectories on the Oxford dataset.** The starting frame is marked by a star. The black and red lines represent the ground truth and the predicted paths, respectively.

Localization Efficiency We further investigate the efficiency of VeLoc under different sampling steps in Table 4. VeLoc achieves strong few-step performance on both datasets. On Oxford, it reaches 3.44m/0.71° with only 2 steps, outperforming DiffLoc even with 20 steps and providing about 4× fewer sampling steps for comparable accuracy. On NCLT, VeLoc achieves 1.02m/2.29° with few steps, while DiffLoc suffers severe degradation at 2 steps and still lags behind in translation accuracy at 20 steps. More results are shown in Appendix.B.2.

Table 4: **Impact of sampling steps.** Errors are reported as translation (m) / rotation (°), and latency denotes the sampling time (ms) per batch. Batch size is set to 28 during evaluation.

Steps	NCLT				Oxford			
	VeLoc(Ours)		DiffLoc		VeLoc(Ours)		DiffLoc	
	Error ↓	Latency ↓	Error ↓	Latency ↓	Error ↓	Latency ↓	Error ↓	Latency ↓
2	1.05 / 2.35	5.80	8.68 / 7.00	4.76	3.44 / 0.71	4.45	13.73 / 3.38	4.59
4	1.02 / 2.29	9.43	1.57 / 2.57	8.59	3.40 / 0.68	7.78	4.04 / 0.85	8.86
8	1.02 / 2.29	18.13	1.30 / 2.35	16.73	3.42 / 0.68	15.41	3.55 / 0.75	16.88
10	1.03 / 2.30	21.04	1.26 / 2.33	20.88	3.42 / 0.66	17.99	3.53 / 0.72	21.12
20	1.04 / 2.32	40.83	1.19 / 2.31	40.68	3.44 / 0.68	34.87	3.49 / 0.72	41.75

4.3 Uncertainty Estimation Evaluation

Failure detection for safe deployment. Detecting localization failures is indispensable for safe deployment. We evaluate failure detection capability using error-rejection curves by progressively excluding lowest confidence score samples from VeLoc and DiffLoc to validate the model-agnostic characteristic as shown in Figure 6. We also adopt variance, which is computed from multiple samples under fixed model parameters, as well as MC dropout as baselines. However, we note here that dropout layer is not implemented in many deterministic architectures including VeLoc, rendering MC Dropout sampling inapplicable. We adopt a post-hoc dropout approximation by injecting a dropout layer at test-time following [66]. More details are shown in Appendix.B.3.

As illustrated in Figure 6, both translation and rotation errors decrease as the rejection rate increases. This trend validates that our SPAR effectively captures the uncertainty and serves as a reliable

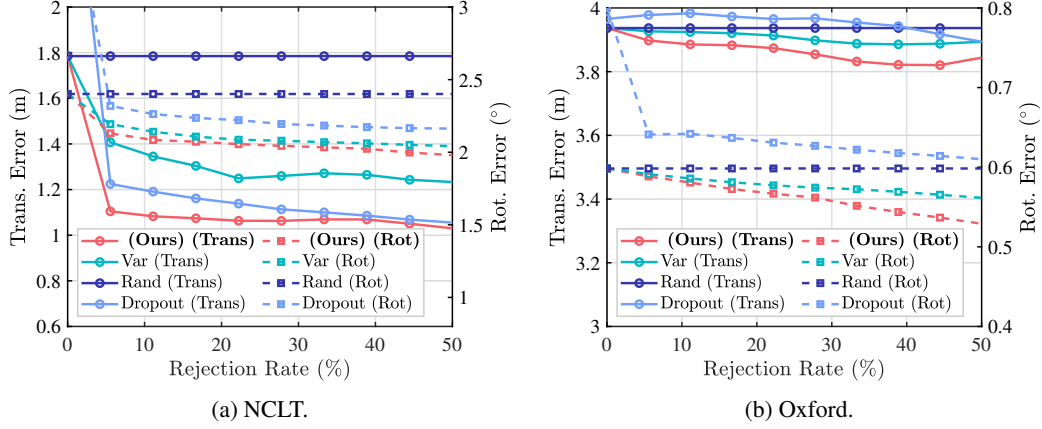


Figure 6: **Error rejection curves on VeLoc.** The solid lines and dashed lines represent translation (left Y-axis) and rotation (right Y-axis) errors, respectively.

indicator of localization quality. Crucially, SPAR consistently outperforms the variance baseline across all rejection levels. Notably, the steepest error drop occurs within the first 5% of rejection on the test trajectory, which indicates that SPAR is highly efficient at filtering out outliers as shown in Figure 7. See Appendix B.3 for more results.

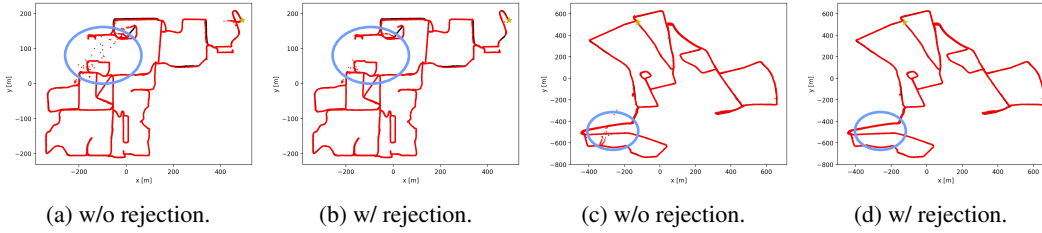


Figure 7: **Visualization of trajectory after rejection.** The results illustrate that our method can effectively filter outliers. We reject the lowest 10% and 2% confident samples for Oxford and NCLT datasets, respectively. The blue circles highlight the significant outliers.

To further validate the generalizability and model-agnostic capabilities of our proposed SPAR module, we extend our evaluation to DiffLoc [13] as shown in Figure 8.

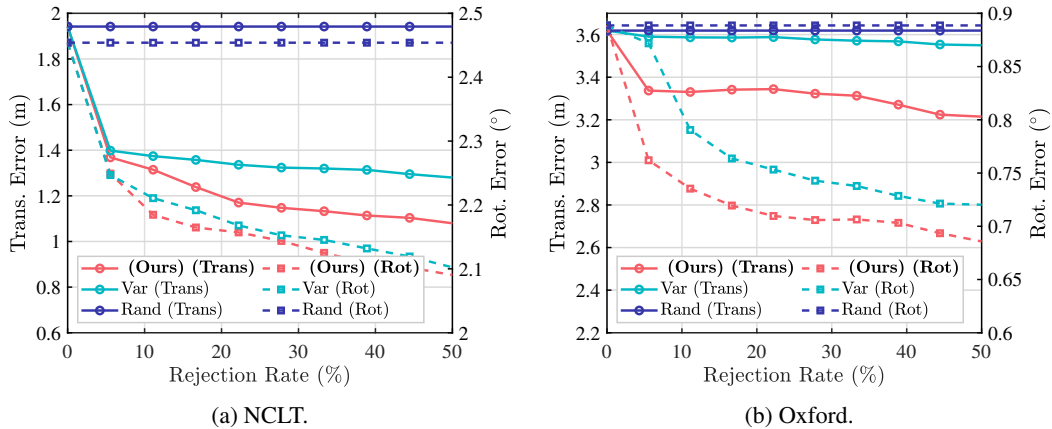


Figure 8: **Error rejection curves on DiffLoc.** The solid lines and dashed lines represent translation (left Y-axis) and rotation (right Y-axis) errors, respectively.

As illustrated in the figure, on the NCLT dataset, localization error decreases progressively as the rejection rate increases. Moreover, the improvement magnitude achieved by the SPAR module surpasses that of the baseline using variance as an uncertainty indicator. However, on the Oxford

dataset, although a similar overall trend is observed, the translation error does not show a significant decrease. This discrepancy is primarily attributed to the inherent complexity of the Oxford dataset, which presents more challenging scenarios.

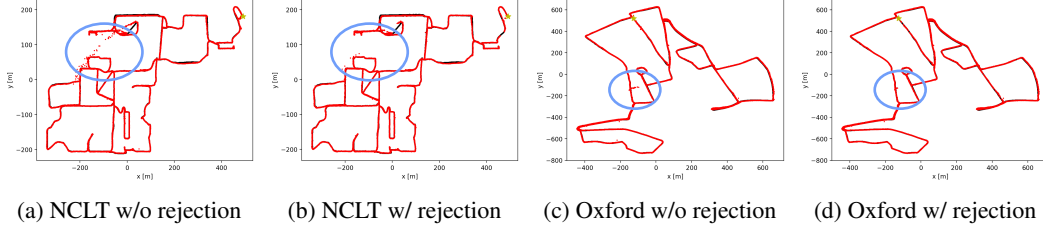


Figure 9: **Visualization of trajectory after rejection.** The results illustrate that our method can effectively filter outliers. We reject the lowest 10% and 2% confident samples for Oxford and NCLT datasets, respectively. The blue circles highlight the significant outliers.

The estimated trajectories by filtering out the samples with the lowest confidence scores are shown in Figure 9. Visualizations of the trajectories confirm that SPAR effectively identifies and excludes geometric outliers.

Score-error alignment. Beyond outlier rejection, confidence scores should correlate negatively with pose error. However, very few works validate this property.

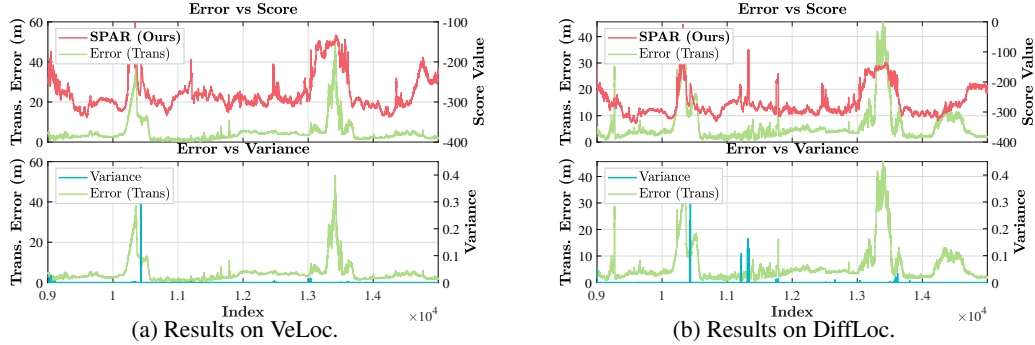


Figure 10: **Pose estimation uncertainty visualization.** We show the relationship between localization error (left y-axis) and variance alongside confidence score estimation result (right y-axis) on the Oxford and NCLT dataset.

To illustrate, we select a representative segment to visualize the correlation between localization error and uncertainty metrics as shown in Figure 10, where confidence scores are negated to align with the error trend. The confidence scores from our SPAR module demonstrate strong negative correlation with translation errors. In contrast, the variance collapses towards zero due to the high determinism of the localization model and thus exhibits weak correlation, failing to scale with the actual error. Our model yields the Pearson coefficient $r_p = -0.6856$ and Spearman’s rank coefficient $r_s = -0.6040$ on NCLT dataset and $r_p = -0.6890$ and $r_s = -0.6161$ on Oxford dataset, indicating a negative correlation. Crucially, our method serves as a model-agnostic, plug-and-play solution that provides universal uncertainty estimation, which we validate with multiple localization methods. Further experiments are shown in Appendix B.4.

5 Conclusion

In this paper, we propose SPAR, a plug-and-play module that reformulates uncertainty estimation as an alignment task via contrastive learning. We further propose VeLoc, a flow-matching framework for LiDAR relocalization, achieving efficient and precise pose estimation. Experiments demonstrate that VeLoc accelerates by $3\times$ compared to the SOTA APR baseline while achieving superior accuracy. Crucially, SPAR exhibits robust and reliable failure detection across multiple localization methods. In summary, our framework achieves a favorable balance between efficiency and accuracy for LiDAR relocalization and realizes reliable failure detection across different localization backbones.

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455 the IEEE/CVF Conference on computer vision and pattern recognition*, pages 11348–11357, 2021.

A Training Setup

A.1 Dataset Details

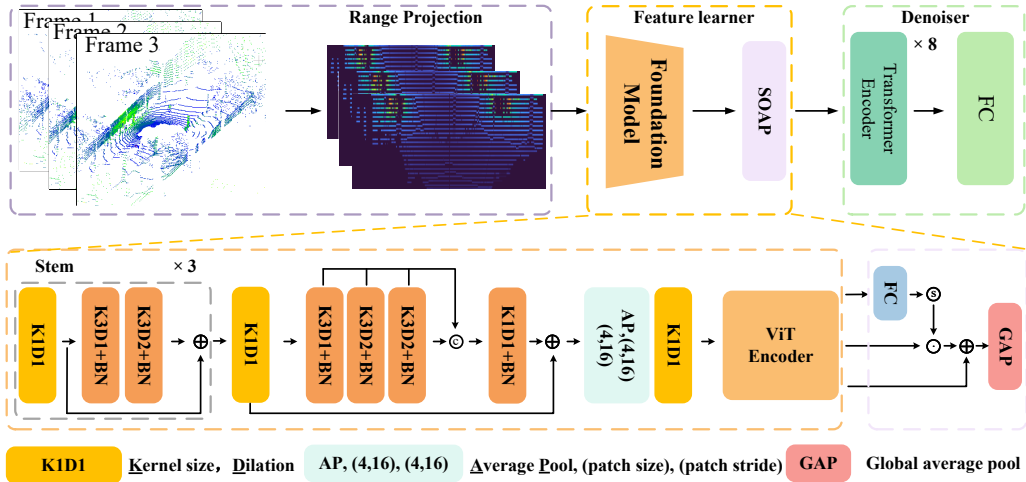
We evaluate our method on two large-scale outdoor benchmark datasets: Oxford Radar RobotCar [63] and NCLT [62]. Both datasets are available online. For each dataset, we list the corresponding data split as shown in Table 5. The Oxford dataset was recorded using an autonomous Nissan LEAF platform navigating for about 10km central route over 32 separate sessions. Our study exclusively utilizes the point cloud data from the left sensor. Similarly, the NCLT dataset was collected at the University of Michigan’s North Campus via a Segway robotic platform. It consists of 27 traversals, each spanning approximately 5.5km, using a single Velodyne HDL-32E LiDAR for spatial scanning.

Table 5: **Dataset sequence details.** We list sequences for Oxford Radar RobotCar and NCLT.

Oxford RobotCar Dataset					NCLT Dataset				
Sequence	Length	Tag	Train	Test	Sequence	Length	Tag	Train	Test
11-14-02-26	9.37km	sunny	✓		2012-01-22	6.1km	overcast	✓	
14-12-05-52	9.22km	overcast	✓		2012-02-02	6.2km	sunny	✓	
14-14-48-55	9.04km	overcast	✓		2012-02-18	6.2km	sunny	✓	
18-15-20-12	9.04km	overcast	✓		2012-05-11	6.0km	sunny	✓	
15-13-06-37	8.85km	overcast		✓	2012-02-12	5.8km	sunny		✓
17-13-26-39	9.02km	sunny		✓	2012-02-19	6.2km	overcast		✓
17-14-03-00	9.02km	sunny		✓	2012-03-31	6.0km	overcast		✓
18-14-14-42	9.04km	overcast		✓	2012-05-26	6.3km	sunny		✓

A.2 Training Details

In our implementation, voxel sizes are set to 0.5 for NCLT and 0.25 for Oxford. The input point cloud sequence is a tuple of size 3 with a spacing of 2 frames. VeLoc is optimized for 150 epochs using AdamW with a peak learning rate of 5×10^{-4} under a one-cycle cosine annealing strategy. The SPAR module adopts a two-stage training scheme over 70 epochs. We initialize the learning rate at 5×10^{-5} for pre-training and 5×10^{-6} for fine-tuning. Both stages employ cosine annealing decay following a 5-epoch linear warmup. The foundation model DINOv2 with ViT-S/16 backbone is used for feature learning.



The framework is shown in Figure 11. For the feature extractor backbone, we replace the patch embedding layer with convolutional stem in the standard ViT. The stem module used processes the

475 $5 \times H \times W$ input range image through a series of residual stages. Initially, three residual layers
 476 expand the feature dimension to 32, followed by subsequent layers that further increase the channel
 477 depth to 64 while maintaining the original spatial resolution. To satisfy the input constraints of the
 478 ViT, we apply average pooling to downsample the feature maps to a grid size of $[H/P_H] \times [W/P_W]$.
 479 A final 1×1 convolution layer then projects these features into a C_{stem} -dimensional space. In
 480 our implementation, with $[H, W] = [32, 512]$, $[P_H, P_W] = [4, 16]$, and $C_{stem} = 384$, the stem
 481 effectively produces 256 visual tokens. These tokens are then processed by a DINO-based ViT
 482 backbone to extract the deep feature map $F \in \mathbb{R}^{256 \times 384}$, excluding the classification token and the
 483 final output equals $F_P \in \mathbb{R}^{384}$. For the sampler, the Transformer consists of 8 encoder layers with
 484 4 attention heads for feature aggregation. The latent embedding dimension is 512. The final fully
 485 connected layers of the sampler are configured as $[512, 64, 6]$.

486 The geometric ADD metric[57], which quantifies the average point-wise displacement induced by
 487 the pose discrepancy, is calculated as:

$$\mathcal{L}_{\text{ADD}}(P_{\text{pred}}, P_{\text{gt}}; \mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{\mathbf{x} \in \mathcal{S}} \|(\mathbf{R}_{\text{gt}}\mathbf{x} + \mathbf{t}_{\text{gt}}) - (\mathbf{R}_{\text{pred}}\mathbf{x} + \mathbf{t}_{\text{pred}})\|_2, \quad (4)$$

488 where $P_{\text{pred}} = [\mathbf{R}_{\text{pred}} | \mathbf{t}_{\text{pred}}]$ and $P_{\text{gt}} = [\mathbf{R}_{\text{gt}} | \mathbf{t}_{\text{gt}}]$ denote the predicted and ground truth poses, respec-
 489 tively, and $\mathcal{S}$ represents the set of 3D points in the scan.

490 A.3 Representation details

491 In VeLoc, the 6-DoF pose is represented as $\mathbf{y} = [\mathbf{t}, \mathbf{r}] \in \mathbb{R}^6$. For the rotation component, we adopt an
 492 axis-angle representation $\mathbf{r}$, derived via the logarithmic map of the unit quaternion. This formulation
 493 maps the $SO(3)$ rotation manifold onto its corresponding tangent space $\mathfrak{so}(3)$. For a unit quaternion
 494 $\mathbf{q} = s + \mathbf{v}\hat{q}$, the logarithmic map is defined as:

$$\log(\mathbf{q}) = 0 + \hat{q} \frac{\mathbf{v}}{|\mathbf{v}|} \arccos(s) \quad (5)$$

495 where the resulting vector denotes the axis-angle representation. During the evaluation phase, the
 496 exponential map is employed to project the regressed vectors back into valid rotation matrices for
 497 rigorous metric computation.

498 Specifically, we enforce a quaternion hemisphere constraint by requiring the real part to be non-
 499 negative, which reduces ambiguity caused by quaternion double coverage. For the identity rotation,
 500 we directly set the logarithm to $[0,0,0]$, thereby avoiding division-by-zero issues. Furthermore, we
 501 examine the target dataset for singular cases. Our task predicts the global pose, including rotation,
 502 rather than relative rotations. On the NCLT 2012-05-26 trajectory, we found no samples with
 503 rotation angles below 0.1° and only 21 samples below 1° , so near-singular cases are extremely rare.
 504 Therefore, although the axis-angle representation is theoretically singular, given our formulation,
 505 numerical preprocessing, and data distribution, we do not observe practical degradation in localization
 506 performance. Moreover, the subsequent optimization is carried out in Euclidean space, which helps
 507 alleviate singularity issues and numerical instability in practice.

508 This representation is advantageous as it bypasses the redundancy and unit-norm constraints inherent
 509 in quaternions, while providing a more continuous and stable space for regression compared to Euler
 510 angles, which are prone to gimbal lock. While $SO(3)$ manifold constraints are not explicitly enforced
 511 during the flow process, this serves as a computationally efficient and effective approximation for
 512 autonomous driving tasks. In such scenarios, the planar motion prior (stable altitude, roll, and pitch)
 513 ensures the numerical stability of the Euclidean approximation. This approach is consistent with all
 514 established baselines in the field. Nevertheless, we acknowledge that direct modeling on the $SO(3)$
 515 Riemannian manifold constitutes a mathematically superior direction, and we aim to integrate such
 516 manifold-aware constraints in future extensions of this work.

517 B Additional Results

518 B.1 Localization Accuracy Comparison

519 We provide the detailed performance on both datasets here, including additional baselines beyond
 520 recent works as shown in Table 6 and Table 7. Consistent with the experimental results presented in

the main text, our method achieves competitive performance on both datasets. More visualizations are shown in Section B.7.

Table 6: **Comparison with the state-of-the-art methods on NCLT dataset.** The mechanism (Mech.) S/A denotes the structure-based localization/absolute pose regression. We highlight the **best** and **second-best** results.

Methods	Mech.	2012-02-12	2012-02-19	2012-03-31	2012-05-26	Average [m°]	Ranks
PNVLAD[4]	S	7.75/6.49	7.47/5.49	6.98/5.67	14.34/7.93	9.14/6.40	12/13
DCP[67]	S	9.84/6.84	8.27/5.16	8.94/5.96	15.62/7.99	10.67/6.49	14/14
SGLoc[6]	S	1.20/3.08	1.20/3.05	1.12/3.28	3.81/4.74	1.83/3.54	4/6
LightLoc[65]	S	0.98 /2.76	0.89 /2.51	0.86 /2.67	3.10/3.26	1.46/2.80	3/3
PointLoc[20]	A	7.23/4.88	6.31/3.89	6.71/4.32	10.02/5.32	7.57/4.60	10/9
PosePN[7]	A	9.45/7.47	6.15/5.05	5.79/5.28	13.47/7.77	8.72/6.39	11/12
PosePN++[7]	A	4.97/3.75	3.68/2.65	4.35/3.38	9.59/4.49	5.65/3.57	8/7
PoseMinkLoc[7]	A	6.24/5.03	4.87/3.94	4.23/4.03	10.32/6.52	6.42/4.88	9/10
PoseSOE[7]	A	13.09/8.05	6.16/4.51	5.24/4.56	12.60/7.67	9.27/6.20	13/11
STCLoc[28]	A	4.91/4.34	3.25/3.10	3.75/4.04	8.67/5.23	5.15/4.18	7/8
NIDALoc[26]	A	4.48/3.59	3.14/2.52	3.67/3.46	6.60/4.56	4.47/3.53	6/5
HypLiLoc[27]	A	1.71/3.56	1.68/2.69	1.52/2.90	2.90/3.47	1.95/3.16	5/4
DiffLoc[13]	A	0.99/ 2.40	0.92/ 2.14	0.98/ 2.27	1.88 / 2.43	1.19 / 2.31	2/2
VeLoc(Ours)	A	0.82 / 2.44	0.73 / 2.12	0.77 / 2.23	1.78 / 2.39	1.02 / 2.29	1/1

Table 7: **Comparison with the baseline methods on the Oxford dataset.** The mechanism (Mech.) S/A denotes the structure-based localization/absolute pose regression. We highlight the **best** and **second-best** results.

Methods	Mech.	15-13-06-37	17-13-26-39	17-14-03-00	18-14-14-42	Average [m°]	Ranks
PNVLAD[4]	S	18.14/3.28	24.57/3.08	19.93/3.13	15.59/2.63	19.56/3.03	14/13
DCP[67]	S	16.04/4.54	16.22/3.56	14.87/3.45	12.97/3.99	15.03/3.89	13/14
SGLoc[6]	S	3.01 /1.91	4.07/2.07	3.37 /1.89	2.12 /1.66	3.14 /1.88	2/8
LightLoc[65]	S	2.33 /1.21	3.19 /1.34	3.11 /1.24	2.05 /1.20	2.67 /1.25	1/4
PointLoc[20]	A	12.42/2.26	13.14/2.50	12.91/1.92	11.31/1.98	12.45/2.17	11/10
PosePN[7]	A	14.32/3.06	16.97/2.49	13.48/2.60	9.14/1.78	13.48/2.48	12/12
PosePN++[7]	A	9.59/1.92	10.66/1.92	9.01/1.51	8.44/1.71	9.43/1.77	9/7
PoseMinkLoc[7]	A	11.20/2.62	14.24/2.42	12.35/2.46	10.06/2.15	11.96/2.41	10/11
PoseSOE[7]	A	7.59/1.94	10.39/2.08	9.21/2.12	7.27/1.87	8.62/2.00	8/9
STCLoc[28]	A	6.93/1.48	7.55/1.23	7.44/1.24	6.13/1.15	7.01/1.28	7/5
NIDALoc[26]	A	5.45/1.40	7.63/1.56	6.68/1.26	4.80/1.18	6.14/1.35	6/6
HypLiLoc[27]	A	6.88/1.09	6.79/1.29	5.82/0.97	3.45/0.84	5.74/1.05	5/3
DiffLoc[13]	A	3.57/ 0.88	3.65 / 0.68	4.03/ 0.70	2.86 / 0.60	3.53 / 0.72	4/2
VeLoc(Ours)	A	3.30/ 0.92	3.76/ 0.62	3.94/ 0.59	2.66/ 0.55	3.42/ 0.66	3/1

Statistical analysis. To provide a more comprehensive evaluation of the performance, we report the statistical results in the following section. Mean and standard deviation (std.) of translation and rotation errors across all test samples for each trajectory are reported in Table 8.

Table 8: **Quantitative comparison of translation and rotation errors.** Mean and standard deviation of translation errors (m) and rotation errors (deg) are reported for DiffLoc and our VeLoc across eight test trajectories.

Method	15-13-06-37	17-13-26-39	17-14-03-00	18-14-14-42	2012-02-12	2012-02-19	2012-03-31	2012-05-26
DiffLoc(Trans)	3.62±3.38	3.63±1.79	4.01±2.31	2.86±1.97	0.99±1.13	0.92±0.90	0.98±3.32	1.88±10.2
VeLoc(Ours)(Trans)	3.30±2.78	3.77±1.61	3.95±1.64	2.67±1.30	0.81±1.06	0.74±1.07	0.77±3.56	1.78±10.8
DiffLoc(Rot)	0.89±3.12	0.69±0.98	0.70±0.51	0.63±1.23	2.40±2.55	2.14±2.09	2.27±2.57	2.43±3.85
VeLoc(Ours)(Rot)	0.92±3.02	0.63±1.16	0.59±0.41	0.56±0.43	2.43±2.62	2.12±2.15	2.23±2.29	2.39±3.52

Statistical results indicate that VeLoc(Ours) achieves more consistent predictions, as evidenced by the lower or equivalent std. across most of the tested trajectories. This stability suggests that VeLoc provides more robust localization compared to the diffusion baseline. Even in specific instances

where our method lags behind slightly, it maintains on-par robustness, demonstrating its reliability in complex environments.

Additionally, we conducted statistical tests to provide further empirical evidence supporting our findings, as detailed in Table 9. We employ the following metrics: Win Rate (%) represents the percentage of total frames where VeLoc achieved a lower localization error than the baseline. Mean Reduc. (m) is the average absolute difference in error between the baseline and VeLoc. p -value (Paired t) is the result of a Paired t-test used to determine if the performance difference is statistically significant. A very small p -value (typically < 0.05) indicates that the improvement is not due to random chance. When computing the p -value, the localization results are sorted by timestamp to ensure that the errors calculated between relative estimates are derived from the same LiDAR scan. All data are measured with the open-source scipy library to avoid confusion.

Table 9: **Statistical significance analysis of translation error reduction.** Win rate, mean reduction of translation error (m), and paired t -test p -values are reported for VeLoc compared with DiffLoc on the NCLT and Oxford datasets.

Dataset	Sequence	Frames	Win Rate (%)	Mean Reduc. (m)	p -value (Paired t)
NCLT	2012-02-12	25,546	63.87	0.1678	1.9×10^{-139}
	2012-02-19	26,543	64.41	0.1917	< 0.001
	2012-03-31	26,205	66.30	0.2064	5.6×10^{-284}
	2012-05-26	26,441	60.47	0.1017	5.8×10^{-4}
	Overall	104,735	63.76	0.1668	Significant
Oxford	15-13-06-37	36,109	55.49	0.3166	< 0.001
	17-13-26-39	42,778	49.65	-0.1335	4.1×10^{-15}
	17-14-03-00	43,705	51.78	0.0580	1.4×10^{-9}
	18-14-14-42	37,109	56.41	0.1927	7.1×10^{-120}
	Overall	159,701	53.13	0.0965	Significant

Table 9 presents a comprehensive statistical analysis of VeLoc’s performance relative to the SOTA APR baseline. On the NCLT dataset, our method achieves an overall win rate of 63.76% and a mean error reduction of 0.1668m. Similarly, on the Oxford dataset, VeLoc maintains competitive performance with a 53.13% win rate. Notably, the p -values across all sequences are substantially lower than 0.001, providing strong empirical evidence that the accuracy improvements of VeLoc are statistically significant and not due to random variation. These results underscore the robustness and efficacy of our proposed method across diverse and large-scale environments.

Likelihood estimation. To evaluate how well the predicted distribution aligns with the true locations, we report negative log likelihood (NLL) as shown in Table 10.

Table 10: **Comparison of NLL calculated under ground truth.** Lower NLL value indicates better distribution approximation.

Dataset	NLL ↓		Error[m/°] ↓	
	VeLoc(Ours)	DiffLoc	VeLoc(Ours)	DiffLoc
NCLT	-8.70	6.87	1.02/2.29	1.19/2.31
Oxford	-8.56	6.95	3.42/0.66	3.53/0.72

The NLL calculated at the GT pose quantifies the alignment between the predicted distribution and the GT pose distribution. VeLoc achieves significantly lower NLL values compared to DiffLoc on both datasets, which suggests that our method provides better distribution approximation.

Performance Analysis. We observe that our method falls back on Oxford although its performance is superior on NCLT dataset. We note that the performance gap on Oxford stems from lower ground truth quality [6], which has a greater impact on our method than on SCR, since SCR employs RANSAC to directly reject GT-induced outliers. Conversely, on NCLT dataset which provides higher GT precision, our method consistently outperforms all baselines, demonstrating superior accuracy. We also note here that the performance is evaluated without uncertainty estimation.

B.2 Localization Latency and Parameter Comparison

In the main text, we compare VeLoc and DiffLoc in terms of sampling-step efficiency. To further examine the practical performance gap, we compare the end-to-end time cost as shown in Table 11. Feature extraction includes the total time for range image and voxel feature extraction. Notably, range image features are reused between VeLoc and SPAR, which significantly reduces the total computational overhead. Iterative optimization is evaluated under 10 sampling steps. SPAR introduces only the latency of pose encoding and feature fusion, resulting in a lightweight and computationally efficient design. The total end-to-end latency remains under 9 ms/sample, which provides a safety margin for real-time deployment. While latency may exhibit marginal fluctuations across trials due to variations in GPU utilization and system overhead, the general efficiency profile remains consistent. Notably, the latency reported for the NCLT dataset in the main manuscript was benchmarked using a two-step inference scheme (NFEs=2), which accounts for the discrepancy with the current results presented here. Latency is measured with a batch size of 28.

Table 11: **Per-sample latency decomposition of VeLoc.** Mean processing time of each component (in ms) and total end-to-end latency are reported for the NCLT and Oxford sequences.

Dataset	Sequence	Feature Extr. (scene)	Feature Extr. (geometry)	Iterative Refinement	SPAR	Total E2E
NCLT	2012-02-12	5.33	1.74	0.68	0.12	7.87
	2012-02-19	5.34	1.67	0.68	0.13	7.82
	2012-03-31	5.31	1.64	0.63	0.13	7.71
	2012-05-26	5.38	1.72	0.67	0.13	7.90
	Mean	5.34	1.69	0.67	0.13	7.83
Oxford	15-13-06-37	5.38	1.14	0.63	0.14	7.29
	17-13-26-39	5.31	1.11	0.64	0.13	7.19
	17-14-03-00	5.32	1.03	0.67	0.13	7.15
	18-14-14-42	5.33	1.12	0.63	0.13	7.21
	Mean	5.34	1.10	0.64	0.13	7.21

In addition, we provide a more comprehensive efficiency comparison, covering FLOPs, MACs, NFEs, performance metrics, and actual latency, as reported in Table 12. We selected different NFE settings to maintain similar performance for fair comparison. As shown in Table 12, VeLoc achieves substantially higher overall inference efficiency.

Table 12: **Comparison of computational complexity and pose accuracy.** FLOPs, MACs, number of function evaluations (NFEs), and mean translation (m) / rotation ($^{\circ}$) errors are reported for VeLoc and DiffLoc.

Method	FLOPs $\downarrow$ (G)	MACs $\downarrow$ (G)	NFEs $\downarrow$	Performance $\uparrow$ (m/ $^{\circ}$)
DiffLoc	50.66	25.33	15	3.51 / 0.72
VeLoc (Ours)	50.2	25.1	2	3.44 / 0.71

B.3 Failure Detection for Safe Deployment

To illustrate the failure detection ability of our proposed SPAR module, we benchmark our method against variance baseline following prior works as shown in Figure 12.

Quantitative Analysis on Heavy-Tailed Distributions. As discussed above, SPAR is not primarily intended to improve the average localization accuracy. Instead, its role is to identify rare but catastrophic localization outliers that dominate operational risk. We therefore perform a stress-test analysis on explicitly selected heavy-tailed sequences from both the Oxford and NCLT datasets. The following sequences exhibit pronounced long-tail behavior and provide a more meaningful testbed for evaluating SPAR.

Metric definitions.

- **P95 and P99 translation errors.** These metrics evaluate the system’s robustness in the long tail by characterizing its worst-case performance. Specifically, the P95 metric indicates the exact error threshold that 95% of the frames fall strictly *below*. In other words, it establishes

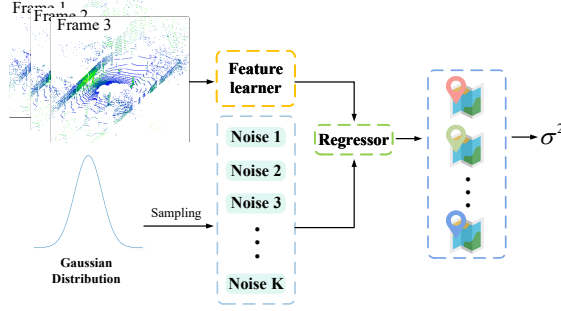


Figure 12: **Uncertainty estimation workflow.** We fix the model parameters and calculate the variance of the generated results conditioned on the same input scan from different noise samples.

Table 13: **Stress testing of SPAR on heavy-tailed sequences.** We report the 95th (P95) and 99th (P99) percentile translation errors to characterize the tail of the error distribution, together with extreme failure detection metrics.

Dataset	Heavy-tailed Sequence	P95 ATE (m)	P99 ATE (m)	AUROC _{10m} ↑	Recall @ 5% Budget ↑
Oxford	15-13-06-37	5.89	15.62	0.907	63.0%
NCLT	2012-05-26	3.53	20.10	0.911	49.4%

an upper bound for the translation drift in the vast majority of operational conditions, revealing how severely the system degrades in the remaining 5% of extreme, challenging scenarios.

- **AUROC for failure detection.** While standard AUROC is designed to evaluate generic binary classifiers, AUROC_{10m} specifically adapts this metric to assess failure detection in a continuous localization task by introducing a physical error threshold. To calculate this metric, we first binarize the continuous ground-truth localization performance. A frame is designated as the positive class, i.e., a failure if its actual translation error exceeds 10m, and as the negative class otherwise. Next, we utilize the system’s estimated uncertainty as the continuous predicted risk score. By systematically sweeping a decision threshold across the full range of these risk scores, we compute the True Positive Rate (TPR) and False Positive Rate (FPR) at each operating point to construct the Receiver Operating Characteristic (ROC) curve. The Area Under this Curve (AUROC_{10m}) mathematically defines the probability that a randomly selected actual failure frame is assigned a higher predicted risk score than a randomly selected nominal frame.

- **Recall @ 5% Budget.** In practical autonomous driving and robotic deployment, continuous false alarms are impermissible. Simply alarming on all frames leads to accurate failure detection but is not allowed in real usage. Constantly triggering computationally expensive multi-sensor fusion or emergency braking would paralyze the system and severely degrade operational availability.

Therefore, we impose an upper bound defined as a 5% intervention budget, which limits the system to evaluate only the TOP 5% of all frames. Under this restrictive formulation, the Recall @ 5% metric evaluates whether the proposed uncertainty estimation module is able to capture all severe trajectory drifts within this limited alarm quota. A high recall demonstrates that the system successfully mitigates false alarms while guaranteeing that catastrophic failures are reliably intercepted.

As shown in Table 13, both datasets exhibit pronounced heavy-tailed behavior, with P99 errors reaching 15.62m on Oxford and 20.10m on NCLT. Under these stress-test conditions, SPAR remains effective at identifying catastrophic failures, achieving AUROC values above 0.90 on both sequences. Moreover, under a strict 5% alarm budget, SPAR successfully captures a substantial portion of the extreme outliers, recovering 63.0% of catastrophic failures on Oxford and 49.4% on NCLT. These results indicate that, while SPAR is not designed as a general-purpose detector for moderate localization errors, it provides strong practical utility in controlling tail risk.

621 **Extended evaluation on NIDALoc.** Similarly, beyond methods mentioned in the main paper, we
622 extend our evaluation to NIDALoc[26] using the same two test trajectories. Given the deterministic
623 nature of NIDALoc and the absence of dropout layers, comparison with the variance-based baseline
624 is excluded. The results are shown in Figure 13 and Figure 14.

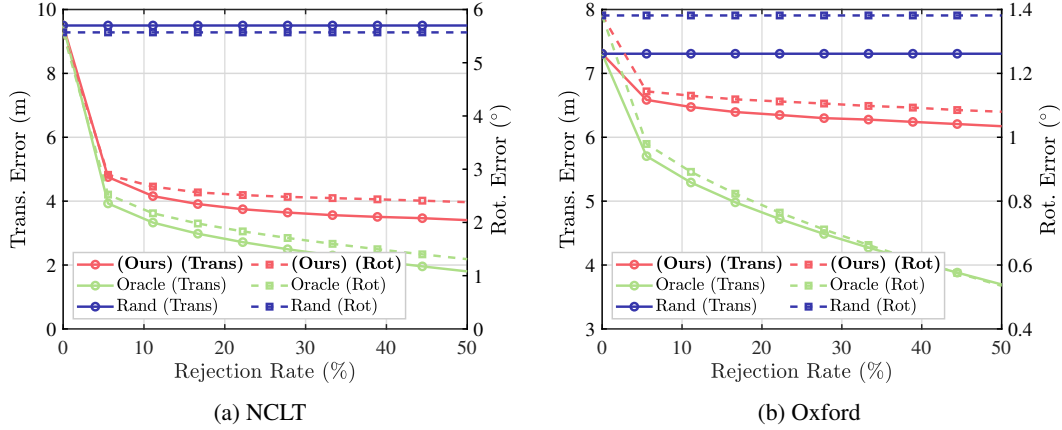


Figure 13: **Error rejection curves on NIDALoc.** The solid lines and dashed lines denote translation (left Y-axis) and rotation (right Y-axis) errors, respectively. We benchmark our method against oracle, variance (ranking by variance), and random (random rejection) on Oxford 15-13-06-37 trajectory and NCLT 2012-05-26 trajectory. Oracle denotes using actual error as confidence score, indicating theoretical lower bound.

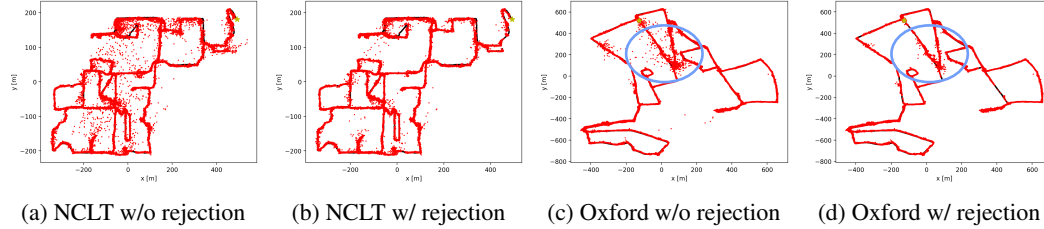


Figure 14: **Trajectories visualization.** The trajectories of NIDALoc before and after the rejection on Oxford 15-13-06-37 trajectory and NCLT 2012-05-26 trajectory are shown.

625 It is evident that the SPAR module effectively identifies and filters out erroneous predictions, thereby
626 enhancing overall localization accuracy. To better compare the performance of our SPAR method
627 under various localization frameworks, we evaluated three distinct localization approaches. The
628 corresponding visualizations are shown in the figure below.

629 **Overall reliability comparison.** Overall, based on further experiments conducted on VeLoc,
630 DiffLoc, and NIDALoc, we validate the generalizability and model-agnostic nature of our proposed
631 module. Based on these findings, we conclude that SPAR functions as a model-agnostic framework.
632 It successfully generalizes across different localization methods by constructing an implicit memory
633 of the spatial structure of the target scene via scan-pose embedding alignment. Our framework
634 offers a versatile tool for enhancing system reliability. The explicit alignment capability enables
635 direct, cross-method benchmarking, allowing SPAR to function as a decision-making supervisor in
636 real-world applications, which can dynamically switch between different localization backbones to
637 prioritize system safety.

638 Crucially, we emphasize that our method serves as a model-agnostic, plug-and-play module that
639 requires no retraining or fine-tuning of the underlying localization backbone. Once trained on the
640 target scene, it serves as an uncertainty estimator. This capability is particularly vital for deterministic
641 architectures lacking stochastic layers (e.g., Dropout or BatchNorm), where conventional Monte Carlo
642 sampling is inapplicable. Unlike alternative solutions that necessitate computationally expensive
643 Deep Ensembles or complex Bayesian parameterization, our framework effectively mitigates these
644 operational complexities.

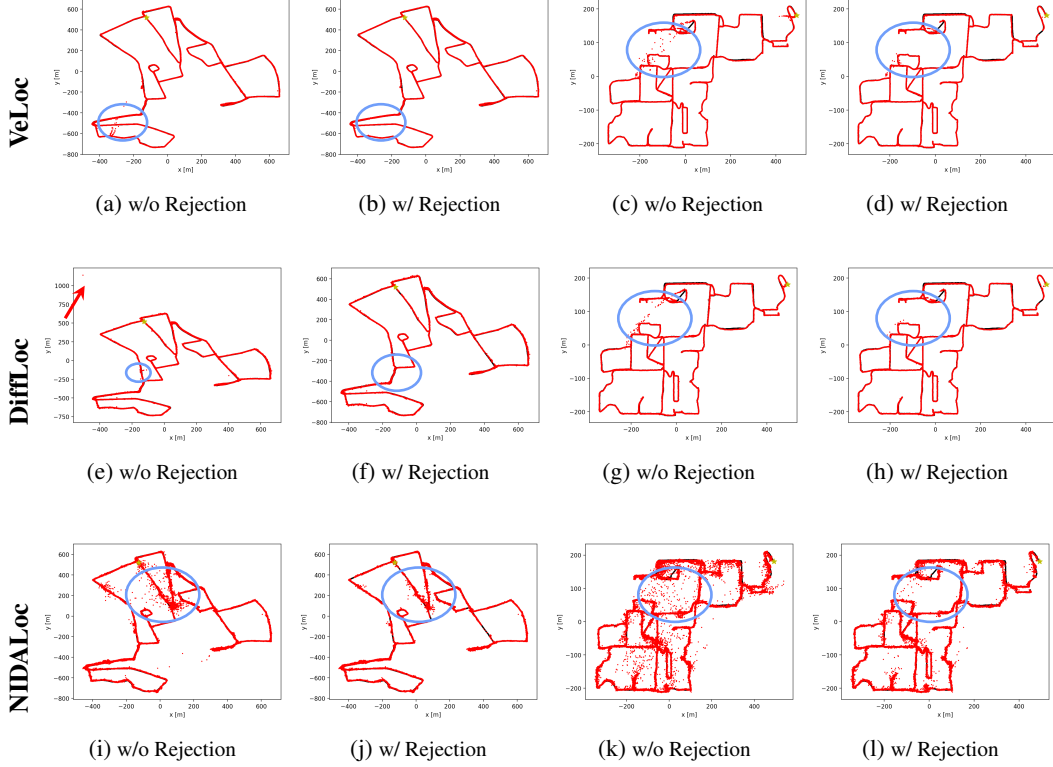


Figure 15: **Trajectory visualization with SPAR module rejection across multiple sequences.** The first column indicates the sequence name. Each row compares the raw trajectory (w/o rejection) against the refined trajectory (w/ rejection) filtered by our post-hoc confidence estimation. Our method successfully detects the failure cases, especially in the circled areas with large errors.

645 **Comparison with other uncertainty estimation methods.** Beyond the comparisons in the main
646 text, we further analyze additional uncertainty estimation approaches. As previously discussed,
647 conventional methods can be categorized as follows.

648 Heuristic methods [10, 45] typically attach task-specific uncertainty heads to a backbone network.
649 These methods are generally not plug-and-play, as the heads take feature maps as input and are
650 jointly optimized with the main task, making them tightly coupled with the localization architecture.
651 Moreover, these heads are often designed under specific assumptions, such as particular pose repre-
652 sentations and error metrics, which further limits their ability to generalize to localization models
653 with different output formats. Consequently, adapting these methods to new models typically requires
654 non-trivial redesign and retraining.

655 Distributional approaches [11, 18, 46] model prediction errors using parametric distributions. Such
656 modeling is often motivated by empirical observations such as histogram fitting, rather than rigorous
657 theoretical guarantees [12]. While errors may appear approximately Gaussian on specific benchmarks,
658 there is generally no formal guarantee that localization errors strictly follow a predefined distributional
659 family under broader conditions. Some methods further adopt evidence regression or predefined
660 parametric forms to improve performance, which may make them less transferable across different
661 models.

662 Meanwhile, variance-based techniques such as MC dropout and deep ensembles require multiple
663 stochastic forward passes, making them computationally expensive for real-time applications. Fur-
664 thermore, MC dropout is not directly applicable to deterministic architectures without dropout layers.
665 To nevertheless provide a competitive baseline, we adopt a post-hoc dropout approximation, which
666 has been shown to be an effective alternative [66]. By injecting a dropout layer at test time with all

other sources of randomness fixed, the variability stems exclusively from the injected dropout. Under this protocol, the variance derived from surrogate dropout substantially underperforms our SPAR module. Moreover, we observe that both the injection position and the dropout ratio still require careful tuning to achieve optimal performance.

Importantly, SPAR enables uncertainty estimation in a single forward pass suitable for real-time deployment, without relying on model ensembles, thereby satisfying both efficiency and plug-and-play requirements. Further visualizations are shown below in Figure 16 and Figure 17.

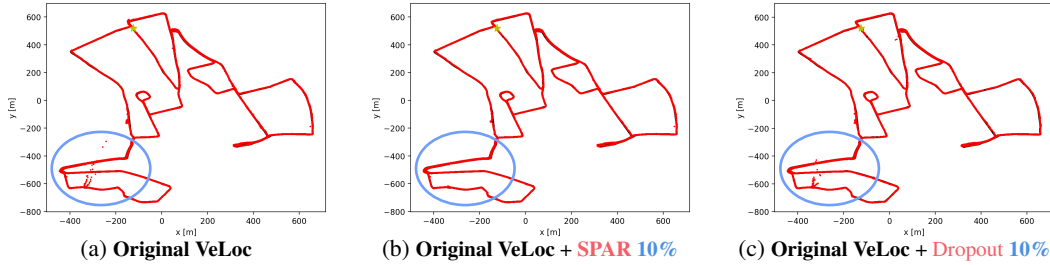


Figure 16: **Visualization on the Oxford dataset.** The starting frame is marked by a star. The black and red lines represent the ground truth and the predicted paths, respectively. The 10% samples with the lowest scores are rejected under different uncertainty estimation methods.

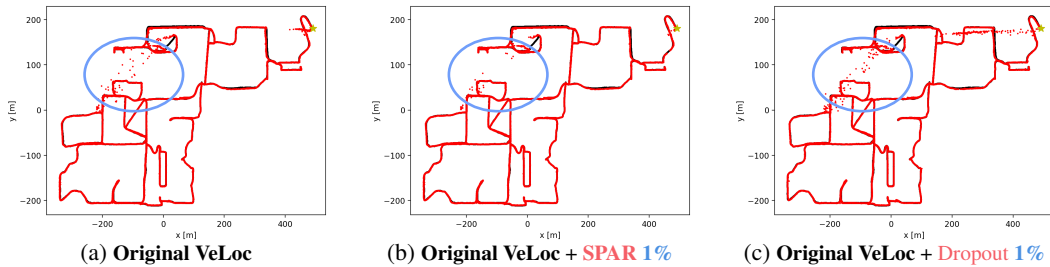


Figure 17: **Visualization on the NCLT dataset.** The starting frame is marked by a star. The black and red lines represent the ground truth and the predicted paths, respectively. The 1% samples with the lowest scores are rejected under different uncertainty estimation methods since our method reaches SOTA on NCLT dataset.

B.4 Score-Error Alignment Verification

As discussed in the main paper, beyond merely rejecting unreliable candidates, a reliable uncertainty estimator should produce confidence scores that are negatively correlated with the pose error. To empirically validate the interpretability and reliability of the proposed scoring mechanism, we visualize the relationship between the predicted scores and the ground truth physical distance errors in Figure 18.

We calculate the pose error following [57] as in the main paper. We correlate the predicted scores with the average point distance error. In both cases, the joint distribution exhibits a negative correlation relationship, where higher confidence scores consistently correspond to lower geometric errors. The linear regression fit (indicated by the red solid line) confirms a negative correlation, suggesting that the learned metric effectively reflects the magnitude of pose error. This property is crucial for practical applications, as it allows for more informed decision-making based on the predicted confidence levels.

B.5 Multi-scale discrimination evaluation.

The SPAR module’s ability to align pose and scan features inspires us to use it for map-database-free place recognition. The overall pipeline is shown in Figure 19.

Additionally, it enables local discrimination for accuracy refinement beyond safe deployment assurance as shown in Table 14. Unlike prior methods[4, 68] which rely on explicit submaps, SPAR

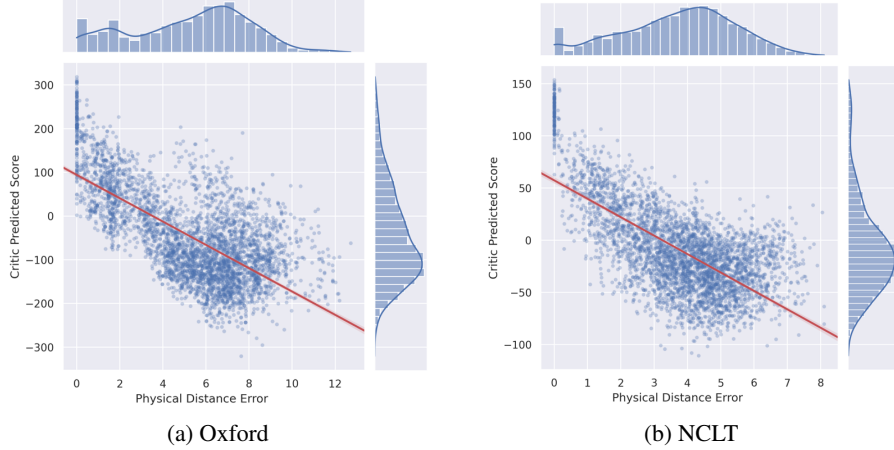


Figure 18: **Correlation visualization.** We visualize the correlation between physical distance error and predicted scores. (a) shows performance on the NCLT dataset, and (b) shows performance on the Oxford dataset. The red solid line indicates the linear regression fit. The marginal density plots reveal the distribution of pose errors and scores.

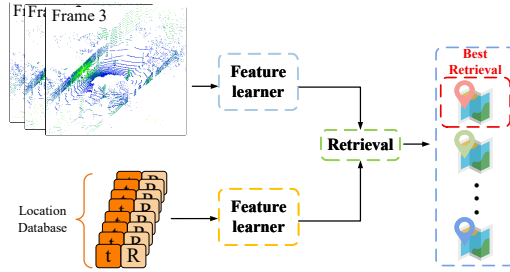


Figure 19: **The overall pipeline of using SPAR for place recognition.** The localization database is extremely lightweight and easy to maintain.

operates on a lightweight pose database. We construct the database using the poses from all training scans and adopt retrieval thresholds of 5m for NCLT and 25m for Oxford[50, 68]. For local discrimination, we generate 19 hard negatives using perturbations of $\sigma_t = 0.05$ and $\sigma_r = 0.02$. Unlike the coarse threshold-based metric used in place recognition, we adopt a stricter metric and directly recall the highest-scoring candidate, which requires distinguishing minimal geometric differences.

Table 14: **Place recognition and local discrimination performance.** We report Recall@k scores of place recognition (Global) and fine-grained verification (Local) on both datasets.

Metric	Oxford		NCLT	
	Global	Local	Global	Local
R@1 (%) ↑	94.74	68.64	91.11	79.70
R@3 (%) ↑	95.47	89.75	97.01	96.89
R@5 (%) ↑	95.90	95.55	98.06	99.10

For place recognition, SPAR achieves robust R@1 scores of 91.11% and 94.74% for NCLT and Oxford datasets, respectively. While our results are slightly inferior to SOTA place recognition methods, we emphasize that our approach requires a lightweight pose database rather than burdensome map database. For fine-grained local discrimination, SPAR achieves a Top-1 Recall of 79.70% on NCLT and 68.64% on Oxford in this challenging task. The high Top-3 Recall shows that SPAR can reliably pick out the correct pose from similar candidates, validating the ability to distinguish minimal geometric differences.

Ablation Study We conduct ablation studies to comprehensively validate the effectiveness of our proposed training strategy as shown in Table 15. The overall results suggest that all components are essential for optimal performance, where the full configuration obtains the optimal performance balance across all metrics.

Table 15: **Ablation study of different loss components.** Best and second-best results are shown.

$\mathcal{L}_{\text{info}}^{\text{inter}}$	$\mathcal{L}_{\text{rank}}^{\text{inter}}$	$\mathcal{L}_{\text{info}}^{\text{intra}}$	$\mathcal{L}_{\text{rank}}^{\text{intra}}$	Oxford			NCLT		
				Global $\uparrow$	Local $\uparrow$	Pearson $\downarrow$	Global $\uparrow$	Local $\uparrow$	Pearson $\downarrow$
✓				75.63	61.17	-0.4480	81.39	70.67	-0.3159
✓	✓			84.17	64.24	-0.5893	85.04	74.47	-0.5347
✓	✓	✓		91.25	67.75	-0.5803	90.40	79.33	-0.4050
✓	✓	✓	✓	94.74	68.64	-0.6890	91.11	79.70	-0.6856

Based on the ablation results presented in Table 15, we validate the effectiveness of the proposed composite loss function on the NCLT and Oxford datasets. We establish a baseline by employing only the inter-scan contrastive loss $\mathcal{L}_{\text{info}}^{\text{inter}}$. While this configuration achieves coarse scan-pose alignment, it exhibits the lowest performance across all metrics. To explicitly supervise the score-error correlation, we incorporate the inter-scan ranking loss $\mathcal{L}_{\text{rank}}^{\text{inter}}$, resulting in significant improvements. Specifically, the Pearson correlation on NCLT improves markedly from -0.3159 to -0.5347, demonstrating that ranking constraints are crucial for aligning confidence scores with pose errors.

To further enhance fine-grained discrimination, we introduce $\mathcal{L}_{\text{info}}^{\text{intra}}$, leading to a substantial gain in retrieval accuracy. Notably, the global recall on the Oxford dataset increases by approximately 7% (reaching 91.25%), indicating that the effective utilization of hard negative mining enhances detail discrimination. Finally, the full model, which integrates the intra-scan ranking loss $\mathcal{L}_{\text{rank}}^{\text{intra}}$, achieves the best performance across all metrics. This configuration yields the most reliable uncertainty estimation, evidenced by the strongest negative Pearson coefficients and high recall rates on both tasks. In summary, these four components are complementary, collectively ensuring both high precision in localization and reliability in uncertainty quantification.

B.6 Discussion and Limitations.

We explicitly acknowledge that while SPAR is effective at capturing catastrophic outliers, it exhibits limited sensitivity to moderate localization drift (e.g., $< 5\text{m}$). This behavior stems from both the characteristics of the training data and a system-level design trade-off.

First, from a data perspective, the spatial resolution of the contrastive alignment is naturally limited. Given the scale of the training trajectories (around 10km per sequence) and the relatively large inter-frame distances typical of vehicle-mounted data collection, capturing fine-grained metric variations is challenging. Furthermore, the inherent ground-truth noise in large-scale datasets such as Oxford further limits the model’s ability to reliably penalize micro-errors during training.

Second, this insensitivity also reflects an important design trade-off in the current formulation. SPAR is inherently more useful on sequences that contain heavy-tailed failures and macro-topological mismatches. Conversely, on highly stable and precise trajectories where extreme errors are rare, its discriminative advantage is naturally reduced, since there are fewer severe mismatches to detect.

In future work, we plan to address this limitation by exploring a hierarchical uncertainty framework. By coupling SPAR’s macro-topological failure detection with traditional micro-level variance estimation, we aim to provide a more comprehensive safeguard that is sensitive to both local trajectory drift and catastrophic structural mismatches.

In addition, we acknowledge that our localization method has not yet achieved the best performance on the Oxford dataset. In future work, we will further optimize the network design, refine the Oxford ground truth [6], and aim to achieve state-of-the-art performance.

B.7 Visualization

In the main text, to ensure a fair performance comparison, we uniformly adopted the optimal sampling step following DiffLoc baseline. However, our VeLoc method can localize within extremely low steps.

746 Furthermore, the performance disparity between DiffLoc and VeLoc becomes even more pronounced
 747 when considering the jitter between adjacent frames.

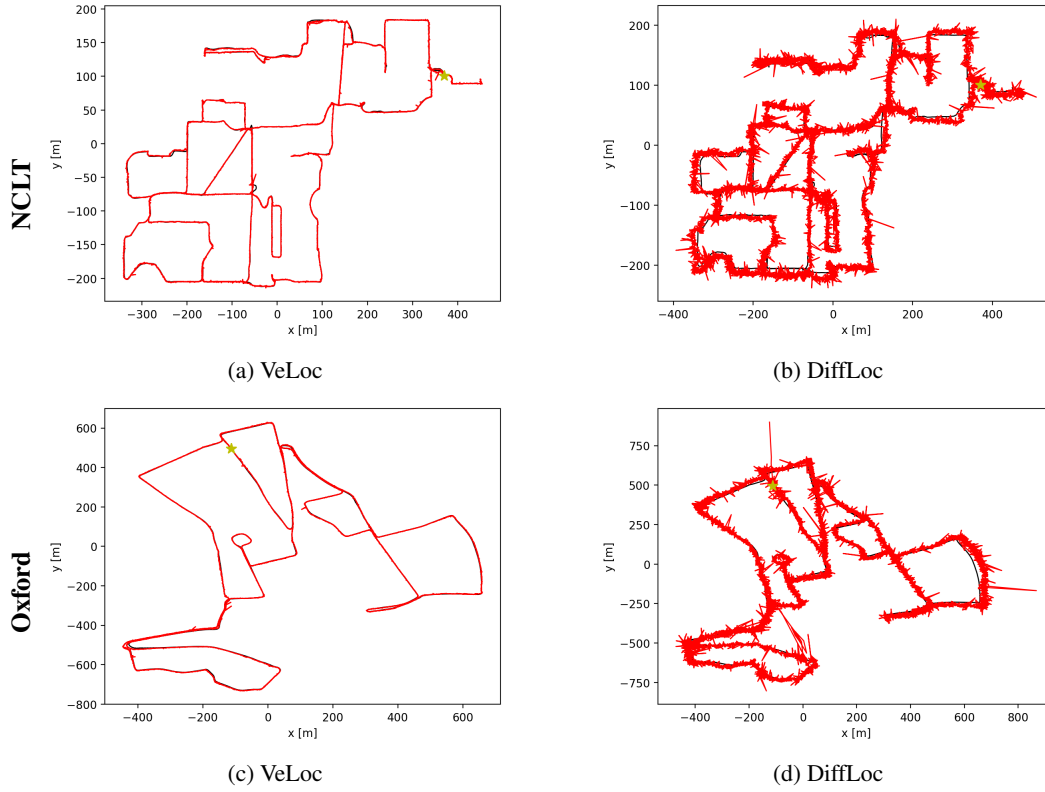


Figure 20: **Visualization on the NCLT and Oxford datasets.** The first row shows results on NCLT, and the second row shows results on Oxford. The starting frame is marked by a star. Both methods are benchmarked under 2 sampling steps. DiffLoc exhibits clear perturbations and drifting around the ground truth.

748 We show more visualization results of NIDALoc, DiffLoc, SGLoc, and the proposed VeLoc in
 749 Figure 21 and Figure 22 on the Oxford and NCLT datasets, respectively. It is important to note that
 750 on certain trajectories, VeLoc fails to achieve the best performance. However, our method achieves
 751 the optimal balance when considering the overall latency and accuracy.

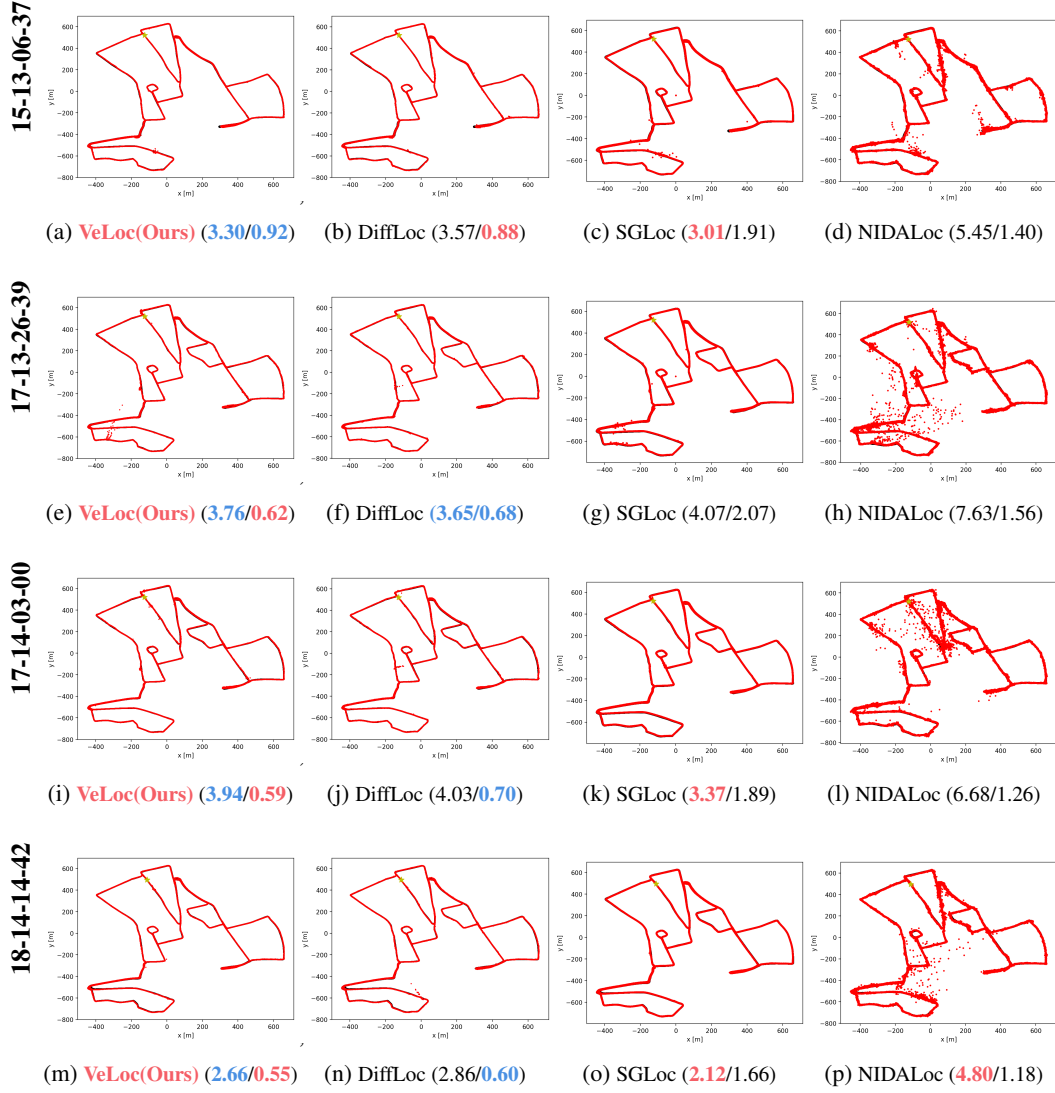


Figure 21: **Visualization results on 4 different Oxford sequences (rows).** The first column indicates the sequence name.

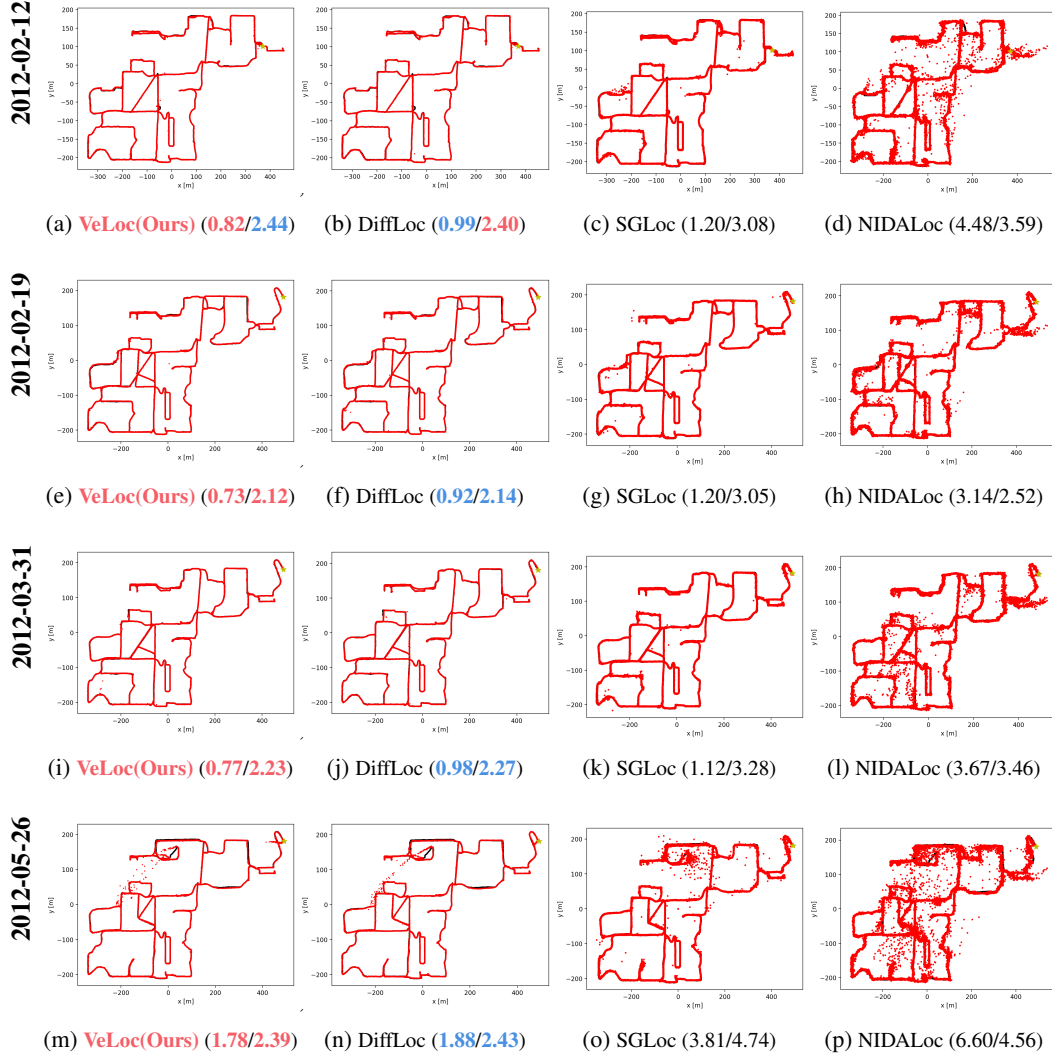


Figure 22: **Visualization results on 4 different NCLT sequences (rows).** The first column indicates the sequence name.

C Likelihood Estimation via Probability Flow

In this section, we provide the theoretical background for evaluating the negative log-likelihood (NLL) of the generated samples. We formulate both flow-matching and diffusion models within the unified framework of the probability flow ODE and the instantaneous change of variables theorem.

C.1 Probability Density Estimation in Continuous Normalizing Flows

The following derivation follows the continuity equation formulation in [41] and is included for completeness.

In sampling-based generative models, the intermediate state $\mathbf{x}_t$ at any time t represents a sample drawn from the marginal distribution $p_t(\mathbf{x})$. Since only empirical samples are accessible, the closed-form density $p_t(\mathbf{x})$ and its score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ cannot be computed directly via differentiation.

To obtain a differentiable trajectory for the time-evolving density, we adopt a fluid-dynamics perspective, treating $p_t(\mathbf{x})$ as a probability fluid flowing through space. Conservation of probability mass implies that the evolution of the density is governed by the continuity equation:

$$\frac{\partial p_t(\mathbf{x})}{\partial t} + \nabla_{\mathbf{x}} \cdot \mathbf{J}(\mathbf{x}, t) = 0, \quad (6)$$

where $\mathbf{J}(\mathbf{x}, t) = p_t(\mathbf{x})\mathbf{v}_t(\mathbf{x})$ is the probability current density, and $\mathbf{v}_t(\mathbf{x})$ is the time-dependent velocity field.

Consider a flow map $\Phi_t(\mathbf{x})$ governed by the ordinary differential equation (ODE) $d\Phi_t(\mathbf{x})/dt = \mathbf{v}_t(\Phi_t(\mathbf{x}))$, with the initial condition $\Phi_0(\mathbf{x}) = \mathbf{x}$. By applying the instantaneous change of variables formula alongside the continuity equation, we derive the rate of change of the log-density:

$$\frac{d}{dt} \log p_t(\Phi_t(\mathbf{x})) = -\operatorname{div} \mathbf{v}_t(\Phi_t(\mathbf{x})). \quad (7)$$

Integrating from time $t = 0$ (data distribution) to $t = 1$ (prior distribution) yields:

$$\log p_1(\Phi_1(\mathbf{x})) = \log p_0(\mathbf{x}) - \int_0^1 \operatorname{div} \mathbf{v}_t(\Phi_t(\mathbf{x})) dt. \quad (8)$$

This allows us to evaluate the data likelihood $\log p_0(\mathbf{x})$ by reversing the integration. In practice, this is implemented by solving an augmented ODE:

$$\frac{d}{dt} \begin{bmatrix} \Phi_t(\mathbf{x}) \\ \log p_t(\Phi_t(\mathbf{x})) \end{bmatrix} = \begin{bmatrix} \mathbf{v}_t(\Phi_t(\mathbf{x})) \\ -\operatorname{div} \mathbf{v}_t(\Phi_t(\mathbf{x})) \end{bmatrix}, \quad (9)$$

initialized with the test sample at $t = 0$. The divergence term $\operatorname{div} \mathbf{v}_t$ is estimated using Hutchinson’s trace estimator for computational efficiency in high-dimensional spaces.

C.2 Probability Density Estimation in Diffusion Models

While diffusion models are typically formulated as stochastic differential equations (SDEs), exact likelihood computation requires a deterministic mapping. Here, we utilize the probability flow ODE formulation [39], ensuring consistency with the standard variance-preserving (VP) SDE schedule [33].

The forward process of a VP-SDE is given by:

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x} dt + \sqrt{\beta(t)} d\mathbf{w}. \quad (10)$$

It is proved [39] that there exists an equivalent deterministic probability flow ODE that shares the same marginal probability densities $\{p_t(\mathbf{x})\}_{t=0}^1$:

$$d\mathbf{x} = \left[-\frac{1}{2}\beta(t)\mathbf{x} - \frac{1}{2}\beta(t)\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \right] dt. \quad (11)$$

783 By substituting the score function approximation $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \approx -\epsilon_{\theta}(\mathbf{x}_t, t)/\sqrt{1 - \bar{\alpha}(t)}$, we identify
 784 an induced velocity field $\tilde{\mathbf{v}}_t(\mathbf{x})$ for the diffusion model:

$$\tilde{\mathbf{v}}_t(\mathbf{x}) = -\frac{1}{2}\beta(t) \left(\mathbf{x} - \frac{\epsilon_{\theta}(\mathbf{x}, t)}{\sqrt{1 - \bar{\alpha}(t)}} \right). \quad (12)$$

785 With this induced velocity field, the NLL for the diffusion model can be computed using the same
 786 augmented ODE method described in Appendix C.1.

787 To ensure the continuous SDE aligns with the discrete-time DDPM trained with a linear schedule, we
 788 define $\beta(t)$ such that the marginal distributions match. The marginal distribution at time t is:

$$\mathbf{x}_t \mid \mathbf{x}_0 \sim \mathcal{N} \left(\sqrt{\bar{\alpha}(t)}\mathbf{x}_0, (1 - \bar{\alpha}(t))\mathbf{I} \right), \quad (13)$$

789 where $\bar{\alpha}(t) = \exp \left(-\int_0^t \beta(s) ds \right)$. In the discrete setting, β_t linearly increases from $\beta_{\min}$ to $\beta_{\max}$.

790 Taking the limit of infinitesimal steps ($T \rightarrow \infty$), the corresponding continuous schedule becomes:

$$\beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min}), \quad t \in [0, 1]. \quad (14)$$

791 This schedule is used to compute the coefficients in the induced velocity field $\tilde{\mathbf{v}}_t(\mathbf{x})$.

792 C.3 Evaluation Metrics

793 Based on the derived log-probability, we evaluate the model using the negative log-likelihood (NLL).
 794 The negative log-likelihood (NLL) measures the alignment between the predicted distribution and the
 795 ground truth:

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log_2 p(\mathbf{p}_i \mid \mathbf{s}_i), \quad (15)$$

796 where $(\mathbf{p}_i, \mathbf{s}_i)$ denotes the i -th ground-truth pose-scan pair in the test set.

797 In the context of probability flow, the NLL measures the concentration of probability mass at the
 798 ground truth location. Minimizing NLL is equivalent to maximizing the likelihood that the generative
 799 trajectory converges exactly to the observed data.

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